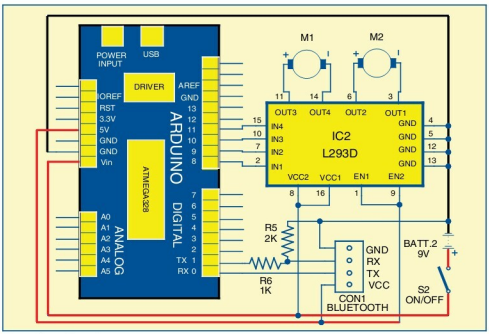


Fig. 3: Controller unit

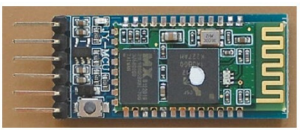


Fig. 4: JY-MCU BT module

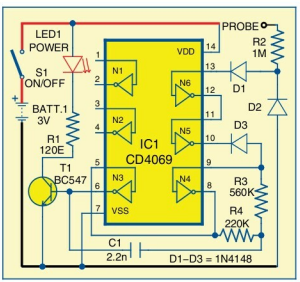


Fig. 5: Detector unit

Pins RX (pin 0) and TX (pin 1) are connected to pins TX and RX of the Bluetooth module, respectively. Pins 5V and GND are connected to pins VCC and GND of the Bluetooth module, respectively. Motor driver IC is connected to pins 8, 9, 10 and 11 of Arduino to drive the motors.

Detector. The detector unit is shown in Fig. 5. It is built around hex inverter CMOS CD4069. Gates N3 and N4 are used as a pulse generator that oscillates around 1000Hz in audio range. Frequency is determined by timing components comprising resistors R3 and R4 and capacitor C1. The test probe is used to sense the presence of 230V AC field around the live wire.

Gates N5 and N6 buffer the weak AC voltage picked up by the test probe. Voltage at output pin 10 of gate N5 can enable or inhibit the oscillator circuit. When the test probe is away from any high-voltage AC field, output pin 10 of gate N5 remains low. As a result, diode D3 conducts and inhibits the oscillator circuit from oscillating.

Simultaneously, output of gate N3 at pin 6 goes low to cut-off transistor T1. As a result, LED1 goes off. When the test probe is moved closer to 230V AC, 50Hz mains live wire, during every positive half cycle, output pin 10 of gate N5 goes high. Thus, during every positive half cycle of the mains frequency, the oscillator circuit is allowed to oscillate at around 1kHz, making the red LED (LED1) blink. (Due to the persistence of vision, LED1 appears to be glowing continuously.) This type of blinking reduces current consumption from the power supply.

The detector section can be detached from the robot and used as a standalone broken-wire-detection device.

Using this gadget, you can also detect fused filament bulbs powered by 230V AC mains or trace the AC mains concealed in the wall or underground cable.

A 3V DC supply is sufficient for powering the detector circuit. You can also provide a 3.3V from Arduino board. AG13 or LR44 type button cells, used inside laser pointers or in LED based continuity testers, can also be used for the circuit. The circuit consumes 3mA during sensing of the AC mains voltage. Alternatively, you can use two 1.5V R6 or AA type batteries.

Software

The software for the robot is written in Arduino programming language.

PARTS LIST

Semiconductors:
IC1 - CD4069 inverter
IC2 - L293D motor driver
T1 - BC547 npn transistor
LED1 - 5mm LED
D1-D3 - 1N4148 signal diode
Bluetooth - JY MCU BT Bluetooth module
Arduino - Arduino UNO board

Resistors (all 1/4-watt, ±5% carbon):
R1 - 120-ohm
R2 - 1-mega-ohm
R3 - 560-kilo-ohm
R4 - 220-kilo-ohm
R5 - 2-kilo-ohm
R6 - 1-kilo-ohm

Capacitor:
C1 - 2.2nF ceramic

Miscellaneous:
M1, M2 - 9V, 20rpm DC motor
BATT.1 - 3V battery (optional)
BATT.2 - 9V battery
S1, S2 - On/off toggle switch
Probe - 10cm single-strand wire antenna
CON1 - 4-pin berg strip female connector
- Chasis and wheels for the robot